

Agricultural Analytics & Climate Forecasting Pipeline

1. Project Overview

This project analyzes agriculture data for locations in South India from 2004 to 2019. The goal is to see how rainfall, temperature, humidity and crop yield change by year, season, crop and location, so it is easier to spot which crops, seasons and places perform best.

The work was done in three stages: AWS S3 to store the raw CSV file, Snowflake (SQL) to load and prepare the data, and a 4-page Power BI report for visual reporting.

2. Dataset Summary

- Columns: 12
- Time Period: 2004 to 2019
- Locations: several places in South India, including Bangalore, Mysuru, Kodagu, Madikeri, Hassan, Mangalore, Chikmangaluru, Davangere, Raichur, Gulbarga and Kasaragodu
- Seasons: 3 (Rabi, Kharif, Zaid)
- Crops: include paddy, cotton, coconut, tea, coffee, ginger, pepper, arecanut, cardamom, cashew, cocoa, groundnut and blackgram
- Rainfall values in the prepared table range from 255 to 4103.

Column	What it holds
Year, Location, Season	When and where the record is from
Crops	Crop grown
Area	Area under cultivation
Rainfall, Temperature, Humidity	Weather values
Soil_type, Irrigation	Soil type and irrigation type
yeilds, price	Crop yield and price (the yield column is spelled "yeilds" in the source file)

3. Data Loading using AWS S3 and Snowflake

The CSV file was stored in an AWS S3 bucket and loaded into Snowflake:

1. **Storage integration** (PBI_Integration) - links Snowflake to an AWS IAM role and the allowed S3 location, so Snowflake has permission to read the bucket.
2. **External stage** (pbi_stage) - a shortcut inside Snowflake that points to the S3 bucket.
3. **Database, schema and table** - created the database PowerBI, the schema PBI_Data and the table PBI_Dataset with 12 columns.

4. **COPY INTO** - loaded the CSV into PBI_Dataset (comma delimited, header row skipped, rows with errors skipped).
5. **Load check** - counted rows per year to confirm the data loaded for every year.

Result: the raw data now sits in the Snowflake table PBI_Dataset, which is kept unchanged.

4. Data Preparation using SQL

A working copy of the table was made (agriculture) so the raw table stays untouched. Then:

- Adjusted values in the copy: rainfall multiplied by 1.1 and area multiplied by 0.9.
- Added a **Year_Group** column to group the years into three periods.
- Added a **Rainfall_Groups** column to group rainfall into Low, Medium and High.

Year_Group	Years
Y1	2004 to 2009
Y2	2010 to 2015
Y3	2016 to 2019

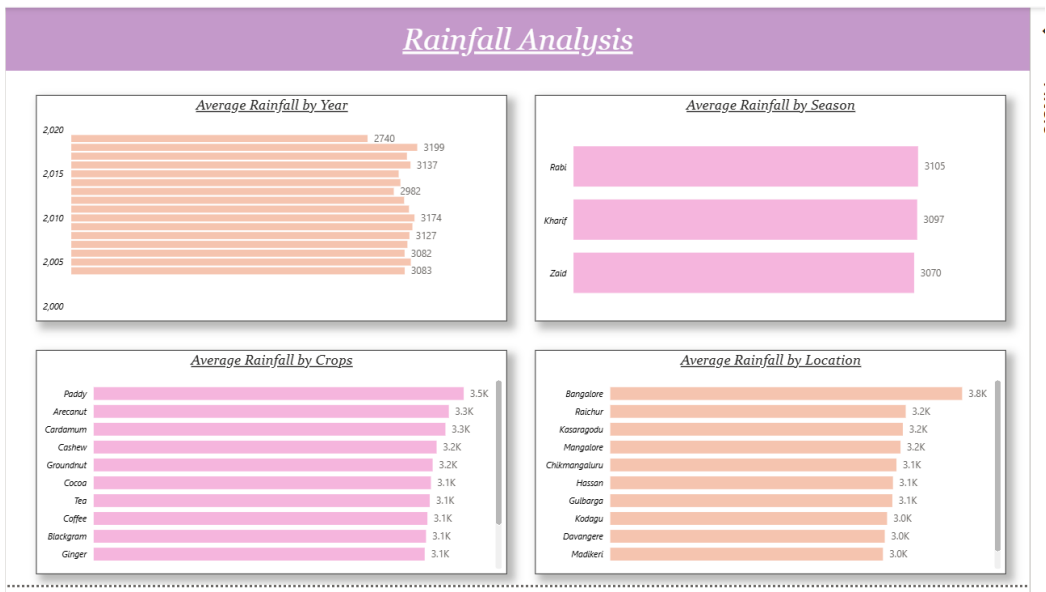
Rainfall_Groups	Rainfall range
Low	255 to below 1200
Medium	1200 to below 2800
High	2800 and above

Result: the agriculture table is ready for Power BI, with the two new grouping columns added.

5. Dashboard in Power BI

Finally, we built a 4-page Power BI report connected to the agriculture table in Snowflake. Every page has four bar charts: average by Year, by Season, by Crop and by Location. The report was then published to Power BI Service.

Page 1: Rainfall Analysis



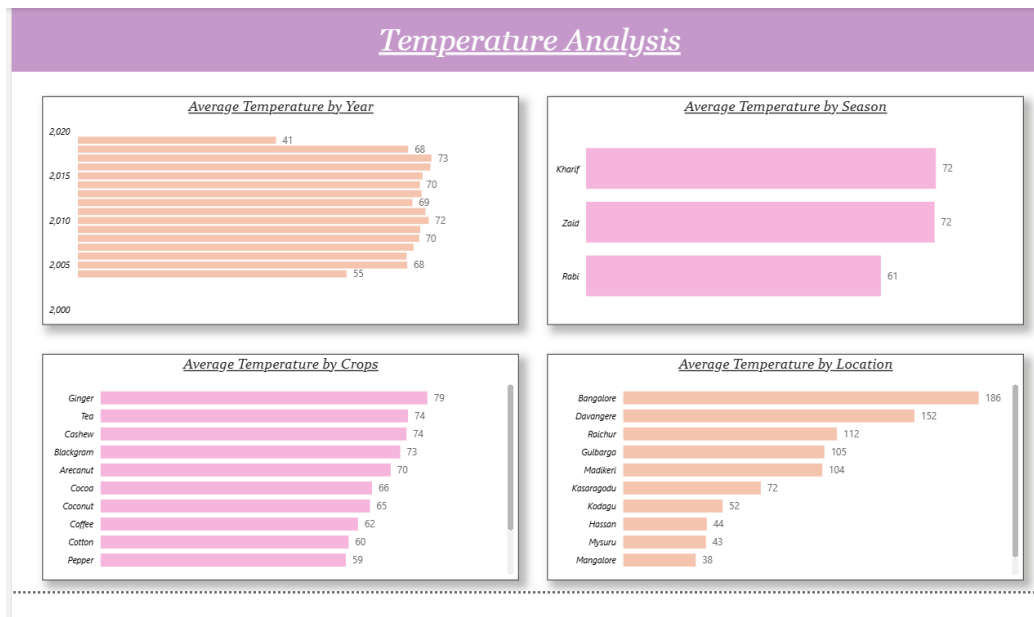
Dashboard: Rainfall Analysis page

Season	Avg Rainfall
Rabi	3,105
Kharif	3,097
Zaid	3,070

Top 5 Crops	Avg Rainfall	Top 5 Locations	Avg Rainfall
Paddy	3.5K	Bangalore	3.8K
Arecanut	3.3K	Raichur	3.2K
Cardamum	3.3K	Kasaragodu	3.2K
Cashew	3.2K	Mangalore	3.2K
Groundnut	3.2K	Chikmangaluru	3.1K

- Bangalore has the highest average rainfall (3.8K). The other locations shown sit between 3.0K and 3.2K.
- Paddy has the highest average rainfall among crops (3.5K). The next crops are at 3.2K to 3.3K.
- The three seasons are almost the same (3,070 to 3,105, a gap of about 1%), so season does not change rainfall much.
- Yearly averages shown on the chart range from 2,740 to 3,199.

Page 2: Temperature Analysis



Dashboard: Temperature Analysis page

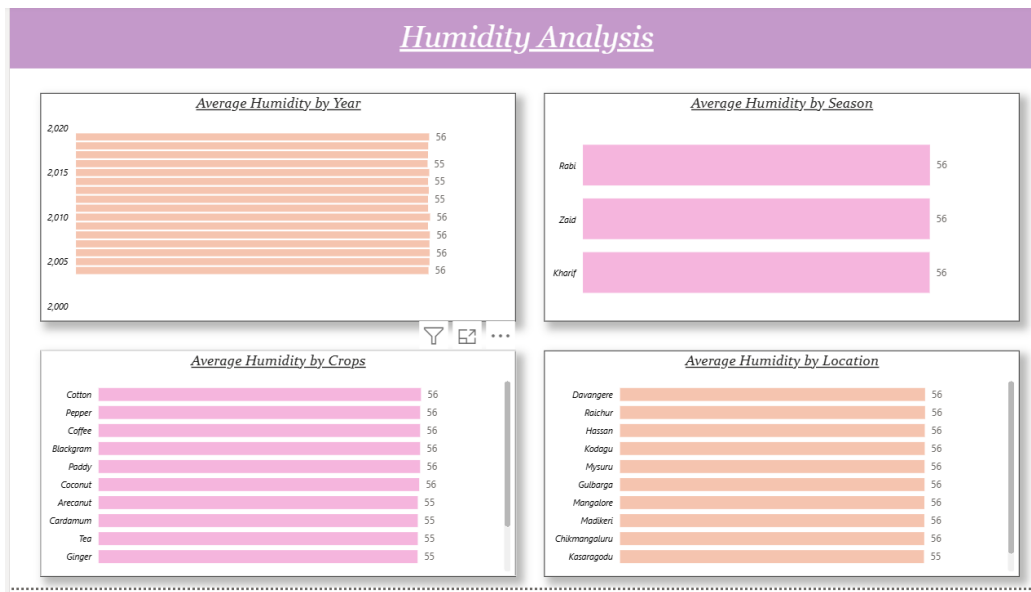
Season	Avg Temperature
Kharif	72
Zaid	72
Rabi	61

Top 5 Crops	Avg Temperature
Ginger	79
Tea	74
Cashew	74
Blackgram	73
Arecanut	70

- Rabi is the cooler season (61), about 11 lower than Kharif and Zaid (72 each).
- Among the crops shown, ginger (79), tea (74) and cashew (74) have the highest averages; pepper (59) and cotton (60) have the lowest.
- Yearly averages shown on the chart range from 41 to 73.

The temperature unit is not stated in the dataset, so values are shown as recorded.

Page 3: Humidity Analysis

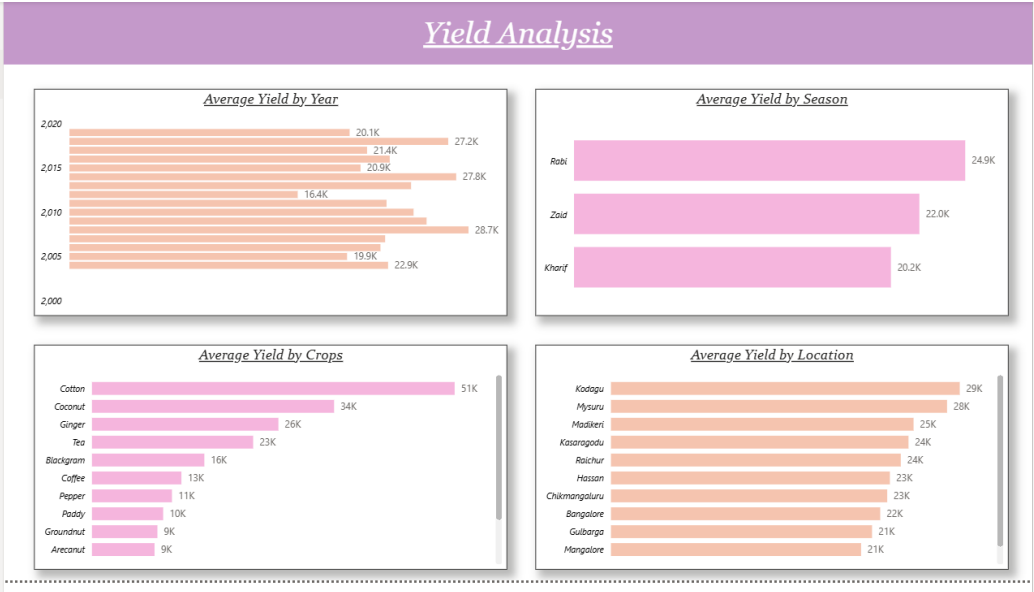


Dashboard: Humidity Analysis page

Season	Avg Humidity
Rabi	56
Zaid	56
Kharif	56

- Average humidity is about 55 to 56 for every year, season, crop and location.
- Cardamum, tea, ginger and arecanut show 55, and the rest show 56, a gap too small to matter.
- Humidity does not separate any group in this data, so it adds little to the analysis.

Page 4: Yield Analysis



Dashboard: Yield Analysis page

Season	Avg Yield
Rabi	24.9K
Zaid	22.0K
Kharif	20.2K

Top 5 Crops	Avg Yield	Top 5 Locations	Avg Yield
Cotton	51K	Kodagu	29K
Coconut	34K	Mysuru	28K
Ginger	26K	Madikeri	25K
Tea	23K	Kasaragodu	24K
Blackgram	16K	Raichur	24K

- Cotton (51K) and coconut (34K) have the highest average yield. Arecanut and groundnut are the lowest crops shown (about 9K).
- Rabi has the highest yield (24.9K), about 23% above Kharif (20.2K).
- Kodagu (29K) and Mysuru (28K) lead by location. Mangalore and Gulbarga are the lowest shown (21K).
- Yearly averages shown on the chart range from 16.4K to 28.7K, so yield changes a lot from year to year.
- Rainfall alone does not explain yield: Bangalore has the highest rainfall (3.8K) but a yield of 22K, while Kodagu has lower rainfall (3.0K) and the highest yield (29K).

6. Key Takeaways

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- **High-yield crops:** cotton and coconut stand out in this data, followed by ginger and tea.
- **Season:** Rabi gives the highest yield and is also the cooler season. This is a pattern in the data, not proof of cause.
- **Location:** Kodagu and Mysuru lead on yield. Bangalore leads on rainfall, but more rain does not mean more yield.
- **Humidity:** it is almost constant everywhere, so it is not useful for telling groups apart.